

Xiaowei Li

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Summary of Expertise

- **Domain Expertise:** Specialized in LLM Post-Training for Tool Use, Function Calling, and Agentic capabilities; expert in LLM Agent Quality and Evaluation.
- **Technical Leadership:** Extensive experience leading teams in building enterprise-grade LLM evaluation benchmarks, agentic orchestration, and simulation/optimization framework.
- **Proven Track Record:** Over 10 years of experience architecting and scaling global-scale distributed services and ML data platforms at Google, Waymo, and Roblox.

Professional Experience

Google, Cloud AI (Gemini & Agent Quality)

Mountain View, CA

Senior Staff Software Engineer & Tech Lead Manager

Oct. 2023–Present

- **Core Contributor** to Gemini 1.0, 1.5, 2.0, 2.5, and 3.0, focusing specifically on function calling, tool use, and agentic capabilities (e.g., MLE, Code Search, etc.)
- Founding engineer for launching **Gemini Function Calling** on Vertex AI and AI Studio.
- Creator of Google's 1st **Enterprise Agentic Evaluation Benchmark**, won the Google Cloud Tech Impact Award.
- Co-initiator and Quality Tech Lead of **Gemini Enterprise** (GCP's flagship agentic platform); launched the core agentic orchestrator and Deep Research agent.
- Technical Lead for **Vertex Agent Builder Quality** (ADK/Agent Engine), delivering advanced agent evaluation (SOTA Autoraters), simulation (User/Environment simulator), and optimization (continuous learning) features.

Roblox

San Mateo, CA

Principal Software Engineer & Tech Lead

May 2022–Sept. 2023

- Architected a new-generation caching control plane designed to handle massive ML inference and non-ML traffic concurrently and elevated end-to-end system reliability to 5 nines.

Waymo (Perception)

Mountain View, CA

Staff Software Engineer & Tech Lead

Oct. 2018–Apr. 2022

- Developed in-house offboard ML models w/ Waymo Research and productionized an Automated Labeling System to generate high-quality perception labels at scale, significantly reducing manual labeling costs (xxx million/yr).
- Designed and built a semantic embedding based Data Mining & Discovery Platform indexing billions of miles of driving logs, enabling rapid discovery of rare events for Perception and Behavior teams (200+ engs).
- Built a highly efficient E2E perception evaluation pipeline to support rapid model and system iterations for 100+ ML engineers.

Google (Ads Data Infra)

Mountain View, CA

Senior Software Engineer

Jun. 2013–Sept. 2018

- Founding engineer of Datahub: co-designed a global-scale metadata management service supporting GCP Data Lake and Google-internal OLAP engines (F1, Dremel) and developed core backend features including query resolution, lineage annotation, and ACL-based governance with millisecond-level latency.
- Built a highly scalable, low-latency payment processing service handling millions of transactions daily for Ads, Google Play, Android Pay, Cloud, etc.
- Developed a resilient ML-based anomaly detection system to identify and mitigate unusual traffic patterns in real-time.

Education

Vanderbilt University

Nashville, TN

Ph.D. in Computer Science & Research Assistant

2008–2013

Affiliated with the NSF TRUST center; conducted research on that uses ML techniques to detect access control flaws and attacks targeting Web Applications and Medical EMR systems